is the study of an organism's anatomy and how it functions (its physiology), as well as how these characteristics relate to the challenges posed by its environment. It shows how the anatomy of an animal or plant is linked to its ability to survive, and to its distribution, abundance, and fertility. Ecophysiology now plays an important role in helping scientists understand how the stresses created by climate change impact on both wild ecosystems and cultivated environments.

"From a physiological viewpoint, freshwater is no more freely available in the sea than in the desert."

Knut Schmidt-Nielsen

Managing temperature

Ecophysiology has revealed a number of specific adaptations for different environments. For example, animals that live in colder regions tend to have larger bodies and smaller legs, ears, and tails than related species living in warmer climes. A larger body has a smaller surface-area-to-mass ratio, and therefore loses heat more slowly, while smaller appendages reduce exposure to frostbite.

In the most extreme cold, the feet of a warm-blooded animal are at risk of becoming frozen to the ground. Mammals in Arctic regions such as musk oxen and polar bears are adapted for life in these conditions by having thick hairs to insulate their feet.

In the Antarctic, the undersides of penguins' feet are insulated by a thick layer of fat. Penguins also have a heat-exchange (or counter-current) mechanism in their legs. The warm blood arriving from the body is cooled to near 32°F (0°C) by the chilled blood arriving from the feet, which warms to body temperature in the process.

Gazelles in Africa use a similar counter-current system to cool their body temperature. They are able to chill the blood entering their head, giving